

OCR Restaurant Bill Benchmark

Lead briefing: evaluation, model decision, and structured extraction pipeline

Executive summary

This project benchmarks four OCR approaches on 10 restaurant bills and converts the selected model's output into structured, validated JSON. PaddleOCR was selected because it achieved the best OCR and business-extraction quality across every accuracy metric while processing all bills without failures.

Selected model: PaddleOCR | Final composite score: 80.5/100 | CPU response time: 47.79 seconds per bill

Project objective

- Compare Tesseract, EasyOCR, PaddleOCR, and EasyOCR plus TrOCR using the same image set and scoring rules.
- Use reviewed ground truth for meaningful receipt text, header fields, and line items.
- Select one model using text quality, structured extraction quality, speed, and reliability.
- Build a local extraction pipeline that produces fields, items, financial values, and validation flags.

Dataset and ground truth

The dataset contains 10 PNG restaurant bills. It includes clean digital receipts as well as photographed and blurred thermal bills. Each bill has one manually reviewed JSON label containing meaningful receipt text, merchant name, bill number, date, GSTIN, payable total, and item name, quantity, and amount. Advertising boilerplate is omitted from the text reference so text metrics reflect information that matters to extraction.

Compared OCR systems

Model	Detection	Recognition
Tesseract	Classical layout analysis	Tesseract LSTM
EasyOCR	CRAFT	CRNN
PaddleOCR	DBNet	SVTR / CRNN
EasyOCR + TrOCR	CRAFT	TrOCR transformer

Evaluation design

All models process the same 10 images. OCR output is grouped into reading-order lines and passed through the same extraction rules, making the comparison fair. The final composite score uses the following weights.

Metric	Weight	Why it matters
CER	25%	Character Error Rate; lower means fewer text-recognition errors.
Token F1	15%	Word overlap; robust when OCR reads lines in a different order.
Field accuracy	35%	Correct merchant, bill number, date, GSTIN, and payable total.
Extracted item F1	20%	One-to-one item name and amount matching; penalizes missed and extra items.
Speed	5%	Mean inference time per bill after model warm-up.

WER is also reported

Word Error Rate is reported for analysis but is not added to the weighted composite because Token F1 already gives a word-level match measure. CER and WER are both lower-is-better; field and item metrics are higher-is-better.

Final benchmark result

Model	Score	CER	WER	Token F1	Field acc.	Item F1	Sec/b ill
PaddleOCR	80.5	0.327	0.474	75.7%	95.7%	93.8%	47.79
EasyOCR + TrOCR	65.9	0.388	0.581	65.7%	78.7%	66.0%	139.84
Tesseract	50.3	0.483	0.710	52.1%	53.2%	29.5%	0.28
EasyOCR	47.3	0.462	0.756	49.4%	48.9%	44.9%	3.95

Why PaddleOCR was chosen

- Lowest CER and WER, so it produced the most accurate raw receipt text.
- Best field accuracy: 95.7%, including perfect total, date, and GSTIN accuracy in this dataset.
- Best parsed line-item F1: 93.8%, with 95.7% recall and 91.8% precision.
- Zero runtime failures across all 10 bills.
- Tesseract was faster, but its 53.2% field accuracy and 29.5% item F1 were insufficient for reliable structured bill extraction.

PaddleOCR extraction pipeline

The selected pipeline is local and command-line based. Run: `python bill_pipeline.py data/images/5.png`. It returns raw OCR evidence, structured business data, and validation status as JSON.

1. Input and preprocessing

The pipeline accepts PNG, JPG, JPEG, BMP, TIFF, and WebP bills. It makes a temporary working copy, converts it to RGB, applies auto-contrast, and applies a modest contrast enhancement. The source image is never modified.

2. Detection and recognition

PaddleOCR detects text regions and recognizes the content of each region. For every region it returns text, bounding box, and confidence. This separates the problem of locating receipt text from reading its characters.

3. Line grouping

Detected word and phrase boxes are grouped by vertical alignment and ordered left-to-right. This turns fragmented detections into logical receipt lines, for example an item name plus its quantity, rate, and amount.

4. Structured extraction

Rule-based extractors identify restaurant name, bill number, date, GSTIN, and payable total. They support common labels such as Bill No, Invoice No, Grand Total, Net Total, Bill Amount, and Total Invoice Value. GSTIN candidates are position-corrected for common OCR substitutions and checksum-checked when possible.

5. Item and financial extraction

Item rows are parsed into name, quantity, rate where available, and amount. Long names split over nearby OCR lines are joined before parsing. Financial extraction keeps subtotal, discount, service charge, VAT, CGST, SGST, IGST, service tax, round-off, tax total, and payable total separate. This prevents confusing a subtotal with the total paid by the customer.

6. Validation

The pipeline returns review flags instead of silently asserting that every result is reliable. It checks for missing total, bill number, and date; invalid GSTIN checksum; no extracted items; low OCR confidence; item sum versus subtotal mismatch; and financial reconciliation mismatch.

Example output meaning

JSON section	Purpose
ocr	Raw grouped lines, line count, and mean OCR confidence. Retains evidence for debugging.
extraction.fields	Merchant, bill number, date, GSTIN, and payable total.
extraction.items	Parsed item name, quantity, rate when identifiable, and amount.
extraction.financials	Subtotal, discount, taxes/charges, round-off, payable total, and tax interpretation.
validation	Whether the output is internally consistent and requires manual review.

Improvements made during the project

- Added reviewed text ground truth for every bill, enabling CER, WER, and Token F1 instead of unavailable metrics.
- Replaced raw OCR-line item credit with parsed-item precision, recall, and F1.
- Added reconstruction of multi-line item names, raising PaddleOCR item F1 from 84.5% to 93.8%.
- Added explicit discount, tax, service-charge, subtotal, round-off, and total fields.
- Added diagnostics to inspect individual bill/model failures and visualization images to separate detection from recognition issues.

Known limitations and next steps

- The sample has 10 bills, so the result demonstrates feasibility rather than broad production generalization.
- Bill 2 loses one bill-number character; bill 6 contains blur/obstruction; bill 9 has an ambiguous brand-versus-branch name.
- CPU-only PaddleOCR averages 47.79 seconds per bill. GPU deployment, batching, and image-cropping strategies are the next speed improvements.
- For broader deployment, extend the dataset with more vendors, layouts, lighting conditions, languages, and handwritten annotations.

Conclusion

PaddleOCR is the best choice for this benchmark because it consistently delivers the strongest raw OCR quality and the strongest structured restaurant-bill extraction. The completed pipeline transforms an image into validated JSON while preserving OCR evidence for review and debugging.